25900683 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Core Course (DSC) - MG3DSCZGY201, Discipline Specific Elective (DSE)-MG3DSEZIM200 - ANIMAL DIVERSITY - NON CHORDATA - II

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer all questions

Each question carries **1** marks

1. The duration of life cycle of silkworm is [K] / [CO1]

   a). 6-8 weeks b). 4-6 weeks

   c). 10-12 weeks d). 25-27 weeks

2. The compound eyes of prawns exhibit which type of vision in dim light? [U] / [CO2]

   a). Mosaic vision b). Superposition image vision

   c). Binocular vision d). Monocular vision

3. Write the characteristic shell shape of Scaphopoda [K] / [CO3]

   a). Spiral b). Bivalved

   c). plate - like d). Tusk-shaped

4. Discuss the significance of Branchiopoda in aquatic ecosystems. [U] / [CO1]

   a). Parasites of fish b). Primary consumers in freshwater food chains

   c). Producers of bioluminescence d). Predators of zooplankton

5. The gills of Fenneropenaeus indicus are located in which part of the body? [K] / [CO2]

a). Abdomen
b). Branchial chamber under the carapace
c). Telson
d). Hepatopancreas

6. Name the marine burrowing organism lived in mucus lined burrows. [K] / [CO3]

a). Sagitta
b). Sipunculus
c). Spirostreptus
d). Sacculina

7. The first larval form of Sacculina is ? [K] / [CO1]

a). Nauplius
b). Cypris
c). Kentrogon
d). Protozoea

8. The ventral plate of a prawn's body segment is called the [K] / [CO2]

a). Carapace
b). Sternum
c). Epimeron
d). Tergum

9. Tube feet in echinodermata are used for: [U] / [CO3]

a). Locomotion and Feeding
b). Reproduction and Anchoring
c). Defense and Protection
d). None of the above

10. The body of trilobites is divided into which regions? [U] / [CO1]

a). Head, thorax, abdomen
b). Cephalon, thorax, pygidium
c). Head and trunk only
d). Prosoma and opisthosoma

11. A farmer notices that his crop yield has reduced even though there are no visible pests. On investigation, it is found that pollinator insects have reduced in the area. What does this indicate? [U] / [CO2]

a). Pests were killed by insecticides
b). Soil fertility has decreased
c). Lack of pollination reduces crop productivity
d). Crops do not require insect pollination

12. Select the nutritional value of edible mollusc [U] / [CO3]

a). High protein and low fat
b). High fat and low protein
c). High carbohydrate and low protein
d). Low essential nutrients

[1x12 = 12]

Part B

Short Answer Questions

Answer any **6** questions

Each question carries **3** marks

13. Comment on any four salient features of Monoplacophora. [K] / [CO3]

14. Categorize the gills in prawn. [U] / [CO2]

15. Report one economic importance of Argulus. [U] / [CO1]

16. Describe why Limulus is called a living fossil. [U] / [CO1]

17. Describe the salient features of Hemichordata. [K] / [CO3]

18. Describe the body symmetry exhibited by adult echinoderms and its evolutionary significance. [U] / [CO3]

19. Mention three ecological advantages of larval metamorphosis in prawn. [K] / [CO2]

20. Name any three larvae of the phylum Mollusca. [K] / [CO3]

[3x6 = 18]

Part C

Short Essay Questions

Answer any **4** questions

Each question carries **5** marks

21. Describe the morphological features of Scolopendra. [K] / [CO1]

22. Discuss the structure and function of thoracic appendages in prawns. [U] / [CO2]

23. Discuss the ecological and economic importance of Holothuroidea. [U] / [CO3]

24. Write an essay on sub phylum Uniramia and its classification with suitable example. [U] / [CO1]

25. Explain the circulatory system in Prawn. [U] / [CO2]

26. Write a short essay on excretory system of Prawn. [U] / [CO2]

[5x4 = 20]